Public Goods and Externality

1 Public Goods

1. Private goods that so far have been the theme of this course possess two properties: rivals and exclusion. Rivals: consumption by an agent reduces the possibility of consumption by other agent (usually to nothing). Exclusion: it is necessary to pay to consume them.

2. (Pure) Public goods: nonrival and nonexclusive. The standard examples: national defense, the police, or emission standards for air quality.

3. Club goods (also known as collective goods or artificially-scarce goods) are a type of good in economics, sometimes classified as a subtype of public goods that are excludable but non-rivalrous, at least until reaching a point where congestion occurs. Examples of club goods would include, cable television, access to copyrighted works, and the services provided by social or religious clubs to their members.

4. Commons are goods which are rivalrous and non-excludable. For example, free parking space, which is a good without exclusion (because it is free) but a rival good (because two cars cannot occupy one space simultaneously).

5. One should not confuse public goods and publicly provided private goods. In many countries there are public educations and public health care programs. Education is not a public good: one can make access to it exclusive by asking that a tuition be paid, and it is a rival good insofar as the cost of educating an extra child is not negligible. The same can be said about health care.

	Excludable	Non-excludable
Rivalrous	Private goods (food, clothing, cars)	Common goods (fish, hunting, water, air)
Non-rivalrous	Club goods (cable TV, patents, cinemas)	Public goods $(free-to-airtelevision)$

1.1 The Optimality Condition

1. Consider an economy where only two goods exists: a private good x and a public good z. The consumer $i = 1, ..., n$ has a utility function $U_i(x_i, z_i)$. The public good is produced from the private good according to $z = f(x)$, where f is increasing and concave. The initial endowment of the economy is X units of the private good.

2. Assume that a quantity x of private good is set aside to produce a quantity z of public good. Then, the scarcity constraint is

$$\sum_{i=1}^{n} x_i \leq X - x$$

3. The consumption of i in public goods is limited only by the total disposable quantity since the public goods is by definition nonrival:

$$\forall\ i = 1, .., n, \quad z_i \leq z$$

4. One obtains the Pareto optima by fixing the utility of the last $n-1$ consumers and by maximizing the utility of $i = 1$ under the feasibility constraint:

$$\max_{\{x_1,..x_n,x;\ z_1,..z_n,z\}} U_1(x_1, z_1)$$

subject to

$$\begin{aligned}
\sum_{i=1}^{n} x_i &\leq X - x \\
z &\leq f(x) \\
\forall\ i &= 1, .., n, \quad z_i \leq z \\
\forall\ i &= 2, .., n,\ U_i(x_i, z_i) \geq \overline{U}_i
\end{aligned}$$

5. Since the utility functions are increasing, one gets $\forall\ i = 1, .., n$, $z_i = z = f(x)$. We will denote g the cost in the private good for production of the public good hereafter:

$$z = f(x) \Longleftrightarrow x = g(z)$$

6. The program is then simplifies to

$$\max_{\{x_1,..x_n,\ z\}} U_1(x_1, z)$$

subject to

$$\forall\ i = 2, .., n,\ U_i(x_i, z) \geq \overline{U}_i$$
$$\sum_{i=1}^{n} x_i \leq X - g(z)$$

7. The Lagrangian can be written

$$L = U_1(x_1, z) + \sum_{i=2}^{n} \lambda_i(U_i(x_i, z) - \overline{U}_i) + \mu(X - g(z) - \sum_{i=1}^{n} x_i)$$

Normalizing $\lambda_1 = 1$ to make the formulas symmetrical, the first order conditions are

$$\sum_{i=1}^{n} \lambda_i \frac{\partial U_i}{\partial z} = \mu g'(z) \tag{1}$$
$$\lambda_i \frac{\partial U_i}{\partial x_i} = \mu, \quad \forall i = 1, ..., n. \tag{2}$$

8. Substituting λ_i in (1) by (2), we have the Pareto-optimal condition which sets the level of public goods production

$$\sum_{i=1}^{n} \frac{\partial U_i/\partial z}{\partial U_i/\partial x_i} = g'(z) = \frac{1}{f'(x)}$$

The expression above is called the Bowen-Lindahl-Samuelson condition (BLS).

9. Notice that
$$\frac{\partial U_i/\partial z}{\partial U_i/\partial x_i} = -\frac{dx_i}{dz}$$
is simply the marginal rate of substitution of consumer i. But as an increase of public good production by definition benefits all consumers, its marginal cost $g'(z)$ must be compared to the sum of all propensities for paying, not to that of a sole consumer as the case would be for a private good. One must equalize marginal cost and the sum of propensity to pay, which is the aim of BLS condition.

1.2 Implementing the Optimum

1.2.1 The subscription equilibrium

1. This solution consists of asking consumers to subscribe part of their wealth to contribute to public good production. Normalize the price of the private good to be 1. Assume that the wealth of consumer i is R_i. He can subscribe s_i to public good production, thus consuming $x_i = R_i - s_i$ private good units. The total quantity of public good produced will be simply $f(\sum_{i=1}^{n} s_i)$.

2. The choice of i of the quantity s_i is made according to the principles of the Nash equilibrium: i will take the quantity subscribed by other consumers s_{-i} as given and will resolve the program

$$\max_{\{x_i, s_i, z\}} U_i(x_i, z)$$

subject to

$$\begin{aligned} x_i + s_i &= R_i \\ z &= f(\sum_{i=1}^{n} s_i) \end{aligned}$$

3. After simplification, the program is

$$\max_{\{s_i\}} U_i(R_i - s_i, f(s_i + \sum_{j \neq i} s_j))$$

This leads to

$$\frac{\partial U_i / \partial z}{\partial U_i / \partial x_i} = \frac{1}{f'(\sum_{i=1}^{n} s_i)}$$

which does not coincide with the optimality condition (BLS.)

4. In effect, when a consumer decides to subscribe to the public good, he takes into account only the increase of his own consumption of public good. In his calculation he neglects the subsequent growth of the utility of all other consumers, so the equilibrium cannot be optimal. The equilibrium leads to a subproduction of public goods.

1.2.2 Voting equilibrium

1. Many procedures consist in asking the agents to vote for their preferred level of public good. Suppose that the public good is produced at a constant marginal cost, $g'(z) = c$ for every z.

2. Each consumer announces a public good level z_i knowing that public good production is $Z(z_1, ..., z_n)$ and the production cost will then be distributed equally among all the consumers. Therefore, each consumer chooses z_i such that

$$\max \ F_i(z_i) = U_i(R_i - \frac{cz_i}{n}, z_i)$$

3. The analysis shows that the voting equilibrium consists of choosing the median agent's preferred level of production. With m as the median agent, the retained level of production is therefore z_m such that $F'_m(z_m) = 0$. So after these immediate substitutions we have

$$\frac{\partial U_m/\partial z}{\partial U_m/\partial x_m}(R_m - \frac{cz_m}{n}, z_m) = \frac{c}{n}$$

4. This result does not coincide with the BLS condition. Note that contrary to the subscription equilibrium, the voting equilibrium does not necessarily lead to a subproduction of the public good.

1.2.3 The Lindahl equilibrium

1. Assume that personalized prices for the public good can be established. Every consumer i must pay p_i per unit of public good that he consumes. The producer of the public good would then perceive a price $p = \sum p_i$ and produce up to the level where his marginal cost equals $p : g'(z) = p$.

2. Every consumer chooses to equate his marginal substitution rate to his personalized price:

$$\frac{\partial U_i/\partial z}{\partial U_i/\partial x_i} = p_i$$

At equilibrium the amount of public good in demand by each consumer must equal the amount of produced, or

$$\forall i = 1, ..., n, \ z_i(p_i^*) = z(p^*)$$

3. From this result it is then deduced that

$$\sum_{i=1}^{n} \frac{\partial U_i/\partial z}{\partial U_i/\partial x_i} = \sum_{i=1}^{n} p_i = p = g'(z)$$

so the BLS condition is verified this time. Now the Lindahl equilibrium leads to a Pareto optimum.

4. The disadvantage to this process is that it assumes as a matter of fact the existence of n "micromarket" upon which a sole consumer buys the public good at his personalized price. In such circumstances it is difficult to maintain the competitive hypothesis unless one can assume that the consumers are divided in homogeneous groups from the point view of their propensity to pay for the public good.

5. It is in every consumer's interest to underestimate his demand, hoping that the others will be more honest and that the level of produced public good will then be high enough to meet his needs; this is the famous free-rider problem.

1.2.4 Personalized Taxation

1. Now suppose that every consumer i is taxed by the state in terms of his consumption z_i of public good: the budget constraint of the consumer i then becomes

$$x_i + t_i(z_i) = R_i$$

2. The consumer then will equate his marginal rate of substitution with the private marginal cost of the public good, whence

$$\frac{\partial U_i/\partial z}{\partial U_i/\partial x_i} = t_i'(z_i).$$

   If the state chooses taxes of the form $t_i(z_i) = p_i^* z_i$, where p_i^* is the personalized price of i in the Lindahl equilibrium, it is clearly that the BLS condition holds.

3. Unfortunately, this operation assumes that the state is privy to very detailed information about the tastes of all consumers, which in general is not true. In practice, the financing of a public good is accomplished by fiscal resources which the state levies on agents (taxes on income, consumption, etc.). These taxes bring on economic distortions and reduces the optimum production level of the public good.

1.2.5 The Pivot mechanism

1. The Vickrey-Clarke-Groves (VCG) mechanism which permits implementation of an optimal social decision in a dominant strategy equilibrium when the utility functions are quasi-linear.

2. Consider an indivisible public good of which 0 or 1 unit can be consumed (e.g., the decision to build a bridge) and for which the unit cost C. The utilities of the agents are assumed to be quasi-linear: if x_i represents the consumption of the sole private good and z that of the public good, we get

$$U_i(x_i, z) = x_i + u_i z$$

where u_i is a parameter of propensity to pay for the public good which is known only by consumer i, who has initial resources R_i in private good at his disposal.

3. The decision to build a bridge brings a benefit of $\sum_{i=1}^{n} u_i$ and a cost of C. In the Pareto optimum the bridge should be built if and only if the sum of propensity to pay exceeds the bridge's cost: $\sum_{i=1}^{n} u_i \geq C$.

4. A first possible mechanism consists of asking consumer to vote on the opportunity of building the bridge, knowing (for example) that each will contribute equally to its financing. Then the consumer i will vote for the construction iff $u_i \geq C/n$. Let F be the cumulative distribution function of the characteristics u_i in the population. The bridge will be constructed iff $F(C/n) \geq 1/2$, that is if C/n does not exceed the median of F.

5. We will present a direct revealing mechanism that implements this optimal decision rule. In terms of auction theory, the first price auction is a direct but nonrevealing mechanism. Vickrey (1961) showed that the second price auction is revealing since it leads consumers' objectives to align themselves on the social objective.

6. Now let us return to the bridge's construction. The VCG mechanism does nothing but generalize the idea of the second price auction. To simplify the notation, we will subtract from the u_i the per capita cost C/n. The new u_i can therefore be either positive or negative, and the Pareto-optimal decision rule amounts to constructing the bridge when $\sum_{i=1}^{n} u_i \geq 0$. Let $d(u)$ be the indicator of that event.

7. Social surplus, with which the utility of each consumer must be identified, is $\sum_{i=1}^{n} u_i d(u)$. Let $v = (v_1, ..., v_n)$ be the statements of the agents. In equilibrium, all consumers must tell the truth; the consumer i therefore evaluates the social surplus at

$$(\sum_{j \neq i} v_j + u_i) d(v)$$

8. If a transfer $t_i(v)$ is deducted from him, we then have (up to a constant)

$$R_i - t_i + u_i d(v) = (\sum_{j \neq i} v_j + u_i) d(v)$$

which implies (still up to a constant)

$$t_i(v) = -\sum_{j \neq i} v_j d(v)$$

In fact, we can even add to the transfer of every agent any quantity independent of his statement.

9. The category of VCG mechanism is therefore characterized by

   (a) $d(v) = 1$, that is, the bridge is constructed if and only if

   $$\sum_{i=1}^{n} v_i \geq 0$$

   (b) the consumer i pays a transfer

   $$t_i(v) = -\sum_{j \neq i} v_j d(v) + h_i(v_{-i})$$

   where $h_i(v_{-i})$ is any sum depending only on statements of the other consumers.

10. Groves and Loeb (1975) prove that all of these mechanism are revealing in dominant strategies. Proof: the utility of i under this mechanism is written

$$R_i + (\sum_{j \neq i} v_j + u_i) d(v) - h_i(v_{-i}).$$

which depends on the statement v_i of i only through $d(v)$. The agent i must have then choose his statement in such a way as to maximize the second term, which gives

$$d(v) = 1 \text{ iff } \sum_{j \neq i} v_j + u_i \geq 0$$

But by definition

$$d(v) = 1 \text{ iff } \sum_{j=1}^{n} v_j \geq 0$$

and $v_i = u_i$ is therefore one of the solutions of the program of agent i, whatever the statements of the other agents may be.

11. Clarke (1971) proposed a particularly interesting VCG mechanism. Choose

$$h_i(v_{-i}) = \max(\sum_{j \neq i} v_j, 0)$$

Then there are four cases to consider:

(a) If $\sum_{j \neq i} v_j > 0$ and $\sum_{j=1}^{n} v_j > 0$, $t_i(v) = 0$

(b) If $\sum_{j \neq i} v_j > 0$ and $\sum_{j=1}^{n} v_j < 0$, $t_i(v) = \sum_{j \neq i} v_j > 0$

(c) If $\sum_{j \neq i} v_j < 0$ and $\sum_{j=1}^{n} v_j > 0$, $t_i(v) = -\sum_{j \neq i} v_j > 0$

(d) If $\sum_{j \neq i} v_j < 0$ and $\sum_{j=1}^{n} v_j < 0$, $t_i(v) = 0$

12. In case (a) and (d), the statement of agent i does not change the decision to build the bridge, and he pays a zero transfer. On the contrary, in case (b) and (c), agent i modifies the decision with his statement, and he then pays a positive transfer; he is called a "pivot" agent. This property gave its name to Clarke's mechanism.

13. It is immediately ascertained that the pivot mechanism is not "balanced": no transfer is negative, and their sum is strictly positive if there is at least one pivot agent. This property unfortunately common to all VCG mechanisms. Therefore, the VCG mechanism do not allow a true Pareto optimum to be attained, since such an optimum requires budget balance as well as efficient decision making.

14. d'Aspremont-Gerard-Varet (1979) showed that if we only ask for implementation in Bayesian equilibrium (and no more in dominant strategies), a fully Pareto-optimal mechanism exists, which then simultaneously ensures efficient decision making and budget balance.

1.2.6 The local vs. international public good

2 Externality

1. There is an external effect, or an externality, when an agent's actions directly influence the choice possibilities of another agent. Pollution, noises and smoke are negative externality. The beekeeper and the nearby orchard is a matter of positive cross-externalities of production. Network effects and scientific research also constitute positive externalities.

2.1 The Pareto Optimum

1. Consider an example comprises two goods, 1 and 2, two firms, and one consumer. Good 1 is supposed to be polluting and good 2 nonpolluting. Firm 1 is situated on river A and produces good 1 from good 2 according to $y_1 = f(x_2)$. The consumer is on river B and has a utility function $U(x_1^c, x_2^c)$. Now firm 2, situated beyond the confluence of the two rivers, produces good 2 from good 1.

2. Since it is downstream from firm 1 (which ejects its waste into river A) and from consumer 1 (who pollutes river B), its production suffers from two negative externalities, so its production function

$$y_2 = g(x_1, y_1, x_1^c)$$

is increasing and concave in its first argument but decreasing in its last two arguments. Denote (ω_1, ω_2) the initial resources of the economy.

3. The Pareto optima of this economy are given by the following program:

$$\max_{\{x_1, x_2, y_1, y_2, x_1^c, x_2^c\}} U(x_1^c, x_2^c)$$

subject to

$$\begin{aligned} x_1^c + x_1 &\leq \omega_1 + y_1 \qquad (\lambda_1) \\ x_2^c + x_2 &\leq \omega_2 + y_2 \qquad (\lambda_2) \\ y_1 &\leq f(x_2) \qquad (\mu_1) \\ y_2 &\leq g(x_1, y_1, x_1^c) \quad (\mu_2) \end{aligned}$$

where λ and μ are multipliers.

4. The first order conditions are

$$\begin{aligned} \frac{\partial U}{\partial x_1^c} - \lambda_1 + \frac{\partial g}{\partial x_1^c}\mu_2 &= 0 \\ \frac{\partial U}{\partial x_2^c} &= \lambda_2 \\ \mu_2 \frac{\partial g}{\partial x_1} &= \lambda_1 \\ \mu_1 \frac{\partial f}{\partial x_2} &= \lambda_2 \\ \mu_1 - \mu_2 \frac{\partial g}{\partial y_1} &= \lambda_1 \\ \mu_2 &= \lambda_2 \end{aligned}$$

5. By eliminating the multipliers, we obtain

$$\frac{(\partial U/\partial x_1^c) + (\partial U/\partial x_2^c)(\partial g/\partial x_1^c)}{\partial U/\partial x_2^c} = \frac{\partial g}{\partial x_1} = \frac{1}{\partial f/\partial x_2} - \frac{\partial g}{\partial y_1}$$

   The left-hand side is the marginal rate of substitution of the consumer, corrected by the fact that his consumption of good 1 entails pollution downstream. This is called the social marginal rate of substitution of the consumer, and it takes into account all of the consequences of his consumption on social welfare.

6. The right-hand side is the marginal rate of transformation of firm 1, corrected by the effect of its pollution on firm 2. As for the central term, this is the usual marginal rate of transformation of firm 2. In effect firm 2 does not pollute, and its social marginal rate of transformation coincides with its private marginal rate of transformation.

7. The principle is the following: in the presence of externality, the usual optimal condition of equality between the marginal rates of substitution of consumers and the marginal rates of transformation of firms bears on the social values of these quantities. The social values take into account the external effects of each agent's decisions on the rest of the economy.

8. In our simplified model, at the optimum, firm 1 must produce less, and the consumer must consume less of good 1 than in the absence of externalities.

2.2 Implementing the Optimum

2.2.1 The competitive equilibrium

1. Assume that firm 2 takes the pollution factors y_1 and x_1^c as given. Then if the prices of goods are p_1 and p_2, the agents' choices will lead to

$$\frac{(\partial U/\partial x_1)}{(\partial U/\partial x_2)} = \frac{p_1}{p_2} = \frac{\partial g}{\partial x_1} = \frac{1}{\partial f/\partial x_2}$$

   which is inefficient.

2. In the competitive equilibrium the agents only take into account the consequences of their choices on their own welfare, in equating the private marginal rates of substitution and of transformation. Under usual conditions, one can show that there is too much of good 1 being produced and consumed.

3. The quantity of fish in the oceans is the archetypal example of a "common resource" that belongs to everyone and to no one. Every fisherman, in pulling a fish from the water, reduces the capacity of the species' reproduction and therefore future stocks. In doing so, he exerts a negative production externality on other fisherman. The competitive equilibrium will indeed exhibit a phenomenon of overfishing.

2.2.2 Quotas

1. The simplest way to arrive at a Pareto optimum is for the government to set quotas specifying that the externality-inducing activities should be set at their optimal level.

2. In the example, it would amount to calculating the Pareto-optimal levels of y_1 and x_1^c, which we denote y_1^* and x_1^{c*}, and to forbid firm 1 from producing more than y_1^* and the consumer from consuming more than x_1^{c*} of good 1.

3. It is nevertheless an oft-adopted solution, under a slightly less brutal form, that consists of limiting the quantity of a certain type of pollutants emitted by firms, or even by consumers (as in carbon emission of automobiles).

2.2.3 Subsidies for depollution

1. It is sometimes possible to install disposition to reduce pollution. Suppose that firm 1 can invest in depolluting a quantity a of good 2 which, without affecting its production, reduces its pollution as if y_1 were $y_1 - d(a)$.

2. First let us look for the Pareto optimum. The scarcity constraint for good 2 becomes

$$x_2^c + x_2 + a \leq \omega_2 + y_2$$

while the production constraint of firm 2 becomes

$$y_2 \leq g[x_1, y_1 - d(a), x_1^c]$$

The quantity a also becomes a maximand of the program.

3. The optimality condition obtained above are unchanged, but that we must add to them a condition to determine the optimal depollution level a^* :

$$d'(a^*) = -\frac{1}{\partial g / \partial y_1}.$$

4. If firm 1 is not prompted to depollute, it will of course choose not to do so. It still seems reasonable for the government to subsidize an expense by paying firm 1 a sum $s(a)$. The profit maximization program for firm 1 then becomes

$$\max_{\{x_2,a\}} [p_1 f(x_2) - p_2 x_2 - p_2 a + s(a)]$$

while firm 2 maximizes in x_1

$$p_2 g[x_1, y_1 - d(a), x_1^c] - p_1 x_1$$

5. The condition of profit maximization of firm 1 implies that

$$s'(a) = p_2$$

So choosing subsidy $s(.)$ such that $s'(a^*) = p_2$ induces the firm to realize the socially optimal expenditure of depollution. Unfortunately, the first-order conditions also entail

$$\frac{\partial g}{\partial x_1} = \frac{1}{\partial f / \partial x_2}$$

So the subsidized equilibrium is still not Pareto optimal.

2.2.4 The rights to pollute

1. Meade (1952) suggested a solution which rests on the finding that externalities contribute to inefficiency only because no other market exists upon which they may be exchanged.

2. He assumes that the state creates a "right to pollute" market: a right to pollute gives the right to a certain number of pollution units. The polluters can buy the corresponding rights to pollute.

3. Thus, firm 1 pays q to firm 2 for each unit of y_1; the consumer pays r to firm 2 for each unit of x_1^c. The consumer's gives

$$\frac{(\partial U / \partial x_1^c)}{(\partial U / \partial x_2^c)} = \frac{p_1 + r}{p_2}$$

while the profit maximization firm 1 implies that

$$\frac{p_1 - q}{p_2} = \frac{1}{\partial f / \partial x_2}.$$

4. As for firm 2, it must take into account payments it receives for determining the number of pollution rights it is ready to sell, it solves therefore

$$\max_{\{x_1,y_1,x_1^c\}} [p_2 g(x_1, y_1, x_1^c) - p_1 x_1 + r x_1^c + q y_1]$$

   By the first-order conditions we obtain

$$\frac{p_1}{p_2} = \frac{\partial g}{\partial x_1}$$
$$\frac{q}{p_2} = -\frac{\partial g}{\partial y_1}$$
$$\frac{r}{p_2} = -\frac{\partial g}{\partial x_1^c}$$

   All of these equalities amount to the optimal condition.

5. The creation of markets for rights to pollution thus implements a Pareto optimum. It is a system far less demanding than the imposition of pollution quotas, since the circulation of such quotas implies that the state knows the preferences and technologies of all agents. It is enough here for the state to open pollution rights markets and let equilibrium establish itself.

2.2.5 Taxation

1. It is conceivable to tax the production of the good 1 at the rate τ and its consumption at the rate t. It is easy to see that if one chooses $\tau = q$ and $t = r$, where q and r are the equilibrium prices on virtual markets of pollution rights, we again find a Pareto optimum.

2. These tax levels are often called Pigovian taxes, in honor of Pigou (1928). The disadvantage to this remedy is that like the imposition of pollution quotas, one needs extraordinarily detailed information on the primitive data of the economy, since the government must be able to calculate the equilibrium prices on pollution rights markets.

2.2.6 The integration of firms

1. For simplicity, we assume that the consumer does not pollute, so $\frac{\partial g}{\partial x_1^c} = 0$. In this case one can envision the two firms are merging. The new firm will then maximize the joint profit as

$$\max_{\{x_1,x_2,y_1\}} [p_1 f(x_2) + p_2 g(x_1, y_1) - p_2 x_2 - p_1 x_1]$$

subject to

$$y_1 \leq f(x_2).$$

2. The first-order conditions give

$$\frac{p_1}{p_2} = \frac{\partial g}{\partial x_1} = \frac{1}{\partial f / \partial x_2} - \frac{\partial g}{\partial y_1}$$

Again we have the Pareto optimum.

2.3 Coase's Theorem

1. In a famous article Coase (1960) doubted the necessity of any government intervention in the presence of externalities. His reasoning is very simple: let $b(q)$ be the benefit that the polluter draws from a level of pollutant production q and $c(q)$ the cost thus imposed on the pollutee.

2. When b is concave and c increasing and convex, the optimal pollution level is given by

$$b'(q^*) = c'(q^*)$$

Suppose that the status quo q_0 corresponding to a situation where $b'(q_0) < c'(q_0)$, and thus the pollution level is too high.

3. Then the polluter and the pollutee have an interest in negotiating. Let ε be a small positive number, and assume that the polluter proposes to lower the pollution level to $(q_0 - \varepsilon)$ against a payment of $t\varepsilon$, where t is comprised between $b'(q_0)$ and $c'(q_0)$. Since $t > b'(q_0)$, this offers raises the polluter's profits; and it is equally beneficial for the pollutee, since $t < c'(q_0)$. Therefore, the two parties will agree to move to a slightly lower pollution level.

4. The reasoning does not stop here so long as $b' < c'$, it is possible to lower the pollution level against a well-chosen transfer from pollutee to polluter. The end result is the optimal pollution level. A very similar argument applies in the case where $b'(q_0) > c'(q_0)$.

5. The Coase Theorem can thus be set forth as follows: If property rights are clearly defined and transaction costs are zero, the parties affected by an externality succeed in eliminating any inefficiency through the simple recourse of negotiation. Stated in this way, the theorem becomes more a tautology: if nothing keeps the parties from negotiating in an optimal manner, they will arrive at a Pareto optimum.

6. In fact Coase (1988) explained that above all he wanted at the time to bring attention to the importance of property rights and of transaction costs. What happens if the hypotheses of the theorem do not hold?

7. First of all note that in many industrial pollution cases, property rights are not defined. For a common resources like ocean fish, it is impossible to identify the polluters and the pollutees and then put a negotiation into place. Even if property rights are clearly defined, transaction costs are rarely negligible. Many literature has above all insisted on the transaction costs due to asymmetrical information (Farrell, 1987).

8. These critiques do not reduce the interest in the Coase theorem. In fact Cheung (1973) shows that in the case of the beekeeper-orchard cross-externality made popular by Meade (1952), there exist in the US contracts between two parties that seek to internalize the externality.